

Functional Requirements

This section describes the functionality of the Task Management System, specifying requirements for the **project board**, **task board**, **collaboration feature**, **notification feature**, and **search feature**.

1. Project Board Requirements

The project board is a digital board that provides a visual of the following parameters: the project's general information, status, condition, phase, potential risks, issues, damage, task dependencies, and progress tracking bar. The Gantt and Burndown charts must also be available on the project board.

- **General project information requirements**

The system must allow project managers to title the project. The system must enable project managers to set the starting date (DD.MM.YYY), deadline (DD.MM.YYY), and estimated date for the project to be finished (DD.MM.YYY). The system must allow project managers to create tasks needed to complete the project and arrange them in order of priority.

- **Project status, condition, and phase and project risk, issue, and damage details requirements**

The system must allow project managers to set the following options for the project's status: Active (the project is being in work), Completed (all project tasks have been completed), Cancelled (the project tasks have not been completed and will not be completed), or On Hold (work on the project tasks have temporarily been suspended).

The system must allow project managers to set the following options for the project's condition: To Be Started (the project has not started yet), On Track (the project is on schedule), At Risk (a risk that may impact the project completion has been identified), Off Track (some damage has been inflicted to the task(s)), and Completed (the project is completed).

The system must allow project managers to set the following options for the project's phase: Initiating (the project's vision, goals, objectives, scope, and potential risks are being defined), Planning (estimated budget, resource plan, and project roadmap are being defined), Executing (the project activities are being performed), Control (necessary corrections are being made), and Completed (the project is completed).

The system must allow project managers to edit the following text fields: Issue (an event that has happened and may impact the project completion), Risk (a potential event that may affect the achievement of the project's completion), and Damage (harm that has happened and impaired the project).

- **Project task dependencies requirements**

The system must allow project managers to set the following task dependency types: Finish to Start (task B cannot be started until Task A is completed), Finish to Finish (Task B cannot be completed until Task A is completed), Start to Start (Task B cannot be started unless Task A has been started), and Start to Finish (Task B cannot be completed unless Task A has been started).

- **Project progress tracking bar requirements**

The system must allow project managers to calculate the percentage of completed tasks on the progress tracking bar. The bar must show 100% only if all the project's tasks have been completed and the Risk and Issue fields are empty.

The system must allow project managers to generate and update the project's Gantt chart visualising not only the project's timeline but almost all the information specified in the project's board (the project's start and due date, progress tracking bar, phase, tasks, and employees working on each task).

The system must allow project managers to generate and update the project's Burndown chart visualising the amount of completed work and the total work remaining.

- **Automated project reports generating requirements**

The system must allow project managers to generate automated reports about the project's current stage including all the above-mentioned parameters.

2. Task Board Requirements

The task board is a digital board that provides a visual of the following parameters: the task's status, start and due date, description, to-do list, tags, comments, and assigned employees. In the following, we use the unifying term *users* for convenience meaning both project managers and project employees.

- **Task title, start date, and deadline requirements**

The system must allow project managers to title the task. The system must enable project managers to set the task's start date (DD.MM.YYY) and deadline (DD.MM.YYY).

- **Task status requirements**

The system must allow users to set the following options for the task's status: Awaiting Assignment (the task is created, but not available; project managers are in the process of assigning the task to the specified employees), In Progress (employees are working on the task), In Review (the task has been completed; the result is being reviewed by the project manager), Completed (the task has been completed; the result has been approved by the project manager), To Be Modified (the task is in the modification process after being rejected by the project manager).

- **Task description and to-do list requirements**

The system must allow project managers to give a textual description of the task. The system must allow project managers to attach images (JPG, PNG, or GIF), videos (MP4, MOV,

WMV, AVI), and text files (DOC, DOCX, TXT, PDF) to the task description. The system must store attached files, which can be achieved by integrating Google Drive.

The system must provide a textual to-do list field, where users could compile a sequence of sub-tasks to be done for the task. A sub-task must be comprised of the title and description fields. The system must allow users to set an order of priority for the sub-tasks. The system must enable users to mark the sub-task after completion.

- **Task assignments**

The system must allow project managers to assign employees to work on the task by mentioning them (@userfirstname_userlastname) in the field Assigned to.

- **Task tags and comments requirements**

The system must allow users to tag (#tasktag) the task.

The system must allow users to post textual comments on the task. The system must also allow them to attach images (JPG, PNG, or GIF), videos (MP4, MOV, WMV, AVI), and text files (DOC, DOCX, TXT, PDF) to the comment. The system must store attached files, which can be achieved by integrating Google Drive. The system must enable users to pin and unpin comments.

- **Task recurrence**

The system must enable project managers to set the task to recur at a specified time regularly.

- **Automated task reports generating requirements**

Users must be able to generate automated reports about the task's current stage including all the above-mentioned parameters.

3. Collaboration Features Requirements

- **General collaboration features requirements**

The system must allow two users to exchange messages via a private chat.

The system must allow a group of users to exchange messages via a group chat. The maximum number of users that can be included in the group chat must not exceed the number of the company's employees.

The system must enable users to share images (JPG, PNG, or GIF), videos (MP4, MOV, WMV, AVI), and text files (DOC, DOCX, TXT, PDF) using both the private chat and group chat applications. The system must store shared files, which can be achieved by integrating Google Drive.

The system must allow users to communicate using video and audio calls and screen sharing, which can be achieved by integrating Zoom.

- **Different time zone collaboration features requirements**

The system must enable users to view other users' time zones by hovering their mouse cursor over the user's name.

The system must allow users to view the default shift hours according to the default time zone.

The system must allow users, working in time zones different from the default one but needing to work at the default shift schedule, to set up a default shift schedule according to their time zone.

The system must allow users, working in time zones different from the default one, to set up their shift schedules according to their time zones.

4. Search and Notifications Requirements

- The system must allow users to search projects or tasks by such attributes as the project's title, starting date, deadline, status, condition, phase and the task's title, starting date, deadline, project employees, status, and tags.
- The system must allow users to receive and create notifications. The system must enable users to set the exact time for notifications (DD.MM.YYY HH:MM:SS). The system must allow users to use the following notification types: sticky notifications (a small-sized visual notification that reminds the user about the task until it is completed) and full-screen notifications (a full-screen-sized visual notification that alerts the user to take immediate action). The system must enable users to snooze notifications. The system must allow users to reschedule notifications.

Non-functional Requirements

This section describes the non-functional requirements for the task management system, which define the system's behaviour, capabilities, and constraints using the following criteria: **usability**, **performance**, **reliability**, **security**, and **compliance**.

1. Usability

Usability assesses the extent to which software facilitates users to utilize its functions easily and appropriately. In the project report, the focus will be done to the following aspects of usability: user interface and accessibility.

The system must have an intuitive user interface with an easy learning curve so that users do not need several hours to master it. To do this, the system must (a) group related functions (e.g., private and group chats and audio and video calls) in one area so that it would be easier to locate them, (b) utilize familiar, recognizable, and descriptive patterns to provide simple, clear navigation and labelling, (c) maintain consistency in interface design by utilizing the same or complimentary fonts and colour combinations, (d) utilize consistent terminology, (e) provide users with the FAQ where answers in concise textual and media forms to the most common questions will be given.

The system must be accessible to users with disabilities complying with the W3C Accessibility Principles (W3C: Web Accessibility Initiative, n.d.).

2. Performance

To describe the performance of the system, the response time and throughput metrics should be considered. For a seamless user experience, the system must have low latency, with response time to the sent request ranging between 200 milliseconds and 1 second. The system must be able to handle a large number of requests without degradation in performance.

3. Reliability

Reliability metrics are measurements used to quantify how dependable and consistent a system is over time (Cheng et al., 2024). The most popular reliability metric for repairable systems is instantaneous availability. The system must either operate properly or be available when requested. Another widely used reliability metric is mean-time-between-failures (MTBF). The system must be of high MTBF with a minimum time of failure. The system must recover quickly after failure and restore all functionalities without data loss. To do this, the system must ensure regular backups.

4. Security

The critical security aspect concerning the task management system is data protection. The system must ensure that user data is encrypted when in transit and at rest. The system must have robust and secure authentication mechanisms and role-based access control.

5. Compliance

The system must comply with the General Data Protection Regulation (GDPR) for data protection and privacy. The system must follow the following key GDPR principles: lawfulness, fairness, transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity, confidentiality, and accountability (GDPR, n.d.). Additionally, the system must ensure that the individuals using the system are aware of how the system's organisation handles both data privacy and data protection.